Q1) Define Poynting Vector

Ans- The Poynting vector represents the rate of energy flow per unit in an electromagnetic

It is given by

$$\vec{S} = \vec{E} \times \vec{H}$$

Where

- $\vec{E}$ = electric field
- $\vec{H}$ = magnetic field

Q2) What are Gradient, Divergence, and Curl?

Ans- These are vector calculus operator used in Physics

1) Gradient ($\nabla \phi$)

- Operates on a scaler field
- Gives direction of maximum rate of change
- Example : Temperature

2) Divergence ($\nabla \cdot A$)

- Measure how much a vector field spreads out from a point
- Positive → source, Negative → sink

3) Curl ($\nabla \times A$)

- Measures rotation or circulation of a vector field
- Used in magnetic and fluid fields

Q3) Describe Dielectric Polarization

Ans- Dielectric polarization is the alignment of electric dipoles in a dielectric material when an external.

- Causes
  - Displacement of charges inside

Types

- Electronic polarization
- Ionic polarization
- Orientation polarization

Q4) Explain Properties of Maxwell's Third Eq.

Ans- Maxwell's Third eq (Gauss Law for Magnetism

$$\nabla \cdot \vec{B} = 0$$

- Properties
- Magnetic field lines are continous loops
- No magnetic monopoles exist

- Indicates magnetic field has no begining or end.

Q5) Define Displacement Current

Ans: Displacement current is the current due to changing electric field, introduced by

- Given by:

$$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$

Where

- ε_0 = Permittivity of free space
- ϕ_E = electric flux